

# Vinay Kamath

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## EDUCATION

### Indiana University

*Masters of Science in Data Science | GPA: 3.8 / 4.0*

Aug 2024 – May 2026

*Bloomington, IN*

### Ramrao Adik Institute of Technology

*Bachelors of Technology in Computer Science, Minor in Data Science | GPA: 9.43/10*

Aug 2020 – May 2024

*Navi Mumbai, India*

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, C, SQL, Java, HTML/CSS, R, LaTeX, Git & GitLab

**Data Science:** TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly, DoWhy, rdrobust

**Cloud & DevOps:** AWS (EC2, S3), GCP, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions)

**AI & Agentic Systems:** RAG, LLM Integration(OpenAI API, Gemini API), Langchain, Langgraph, ChromaDB

**Data Engineering/Backend:** FastAPI, SQL, Data Pipelines, Streamlit

**Visualization & Analytical Tools:** Tableau, Power BI

**Geospatial & Network Analysis:** GeoPandas, OSMnx, QGIS, Google Earth Engine, Gephi

## EXPERIENCE

### Graduate Research Assistant (Geo-spatial Data Science)

Jan 2026 – present

*Luddy School of Informatics & Computing*

*Bloomington, IN*

- Reconstructed missing geospatial data for 65K+ irrigation canal records by implementing probabilistic record linkage and fuzzy matching, improving dataset coverage from 83% → 96%.
- Made a spatial indexing pipeline using R-trees to compute proximity between 60K+ global dams and HydroBASINS Level-7 regions, generating geospatial features and exporting results as GIS-compatible layers (shapefiles) for spatial validation and analysis in QGIS.
- Executed Google Earth Engine workflows (map-reduce) to aggregate 24 years (2001–2024) of MODIS vegetation data (250m resolution) into basin-level features (mean, max, median), generating a dataset across 58K+ regions.
- Engineered features linking dam locations with upstream and downstream HydroBASINS regions, integrating vegetation indices and hydrological boundaries to quantify pre- and post-dam environmental changes.
- Applied causal inference methods (difference-in-differences, propensity score matching, fixed effects regression) on panel data to estimate the impact of dam construction on upstream vs downstream regions, identifying an 18% increase in agricultural output in high-exposure areas.

### Summer Research Assistant

Apr 2025 – Jul 2025

*Indiana University*

*Bloomington, IN*

- Developed a Python pipeline to scrape and process 66,000+ dealership images across 13 U.S. states using the Google Street View API, capturing multi-angle views and filtering for OCR-ready images.
- Integrated Google Cloud Vision API and built a keyword-based classification system to identify “Buy Here Pay Here” dealerships, achieving 72% precision and reducing false positives by 35%.
- Developed a python script to extract and parse dealership websites, enhancing BHPH detection coverage by 18%.
- Optimized runtime of the python script by 42% using hash maps and parallelized logic; maintained full pipeline documentation and reproducible scripts on GitHub.

### Machine Learning & Platform Engineer

Jan 2024 – May 2024

*Physarum Software Private Limited*

*Bangalore, India*

- Automated 6 ML pipelines for fraud detection, churn prediction, and loan risk scoring, processing 5,000+ records per hour using ML models(Logistic Regression, Random Forests, Gradient Boosting, and XGBoost) with scikit-learn and YAML-based configs
- Implemented and deployed five RESTful APIs using FastAPI to serve trained models.
- Containerized 8 ML services with Docker and managed 20+ deployment cycles on Kubernetes.
- Built GitLab CI/CD pipelines and standardized ML projects scaffolding using Copier.
- Deployed production ML models using Google Cloud Run and Vertex AI, automating infrastructure with Terraform.

## PROJECTS

### Multi-Agent Resume Tailoring System | *Langgraph, Langchain, OpenAI, ChromaDB, Streamlit*

- Built a LangGraph-based multi-agent system that analyzes resumes against job descriptions, extracting and evaluating 50+ skills and keywords per application.
- Implemented a RAG pipeline with ChromaDB for vector storage and semantic search, retrieving role-specific recommendations from 500+ ATS keywords and resume templates.
- Developed a Streamlit interface for PDF resume uploads and real-time display of ATS scores, skill gaps and bullet rewrites.